Game Documentation

1. General Information

Game Title: Why are You Running

Student Name: W.M.P Sesath Insandu

IT Number: IT22249920

2. Summary of the Game Concept

The game is a fast-paced, 2D endless runner set in a vibrant cyberpunk city. The player takes control of a character attempting to survive as long as possible while navigating randomly generated platform gaps. Along the way, the player must dodge or destroy robotic enemies using a limited-ammo boomerang combat system. To survive and achieve a high score, the player collects floating health orbs, score coins, and boomerang ammo pickups. The game features progressive difficulty, where the global speed of the environment and obstacles continuously increases the longer the player survives.

3. Description of Core Features

- Endless World Generation: The game utilizes a dynamic PlatformSpawner script that instantiates various platform prefabs at randomized vertical heights and randomized time intervals, creating unpredictable gaps. The platforms themselves move to the left using an ObjectMover script, simulating forward movement.
- Progressive Difficulty: A central GameManager script continuously calculates a globalSpeedMultiplier. This multiplier is applied to the background parallax scrolling, the platform movement speeds, and the spawn timers, ensuring the game becomes proportionally harder over time.
- Collision and Item Collection: The player utilizes a combination of Rigidbody2D and Box/Circle Colliders. An OnTriggerEnter2D function efficiently handles interactions with tagged objects, such as taking damage from Obstacles (Robots), healing from HealthOrbs, restocking the ammo array from AmmoPickups, and adding points via ScoreGates.
- Game State UI: The user interface features dynamic visual icons for both Health and Ammo. It uses arrays of UI Images that toggle on and off to represent the current state. The game also includes a fully functional Pause Menu and Game Over screen that manipulate Time.timeScale to freeze and unfreeze the game state.

4. Control Guide

- Jump / Double Jump: Left Mouse Click OR Spacebar

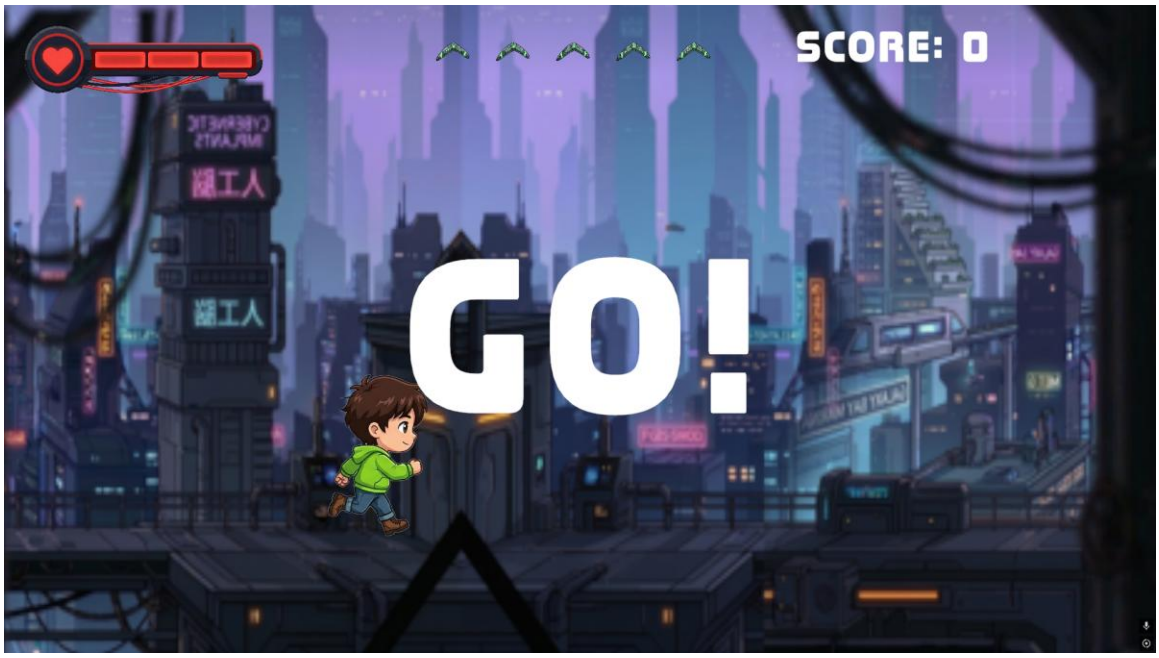
- Throw Boomerang (Attack): Right Mouse Click OR 'F' Key
- Pause Game: Escape (Esc) Key

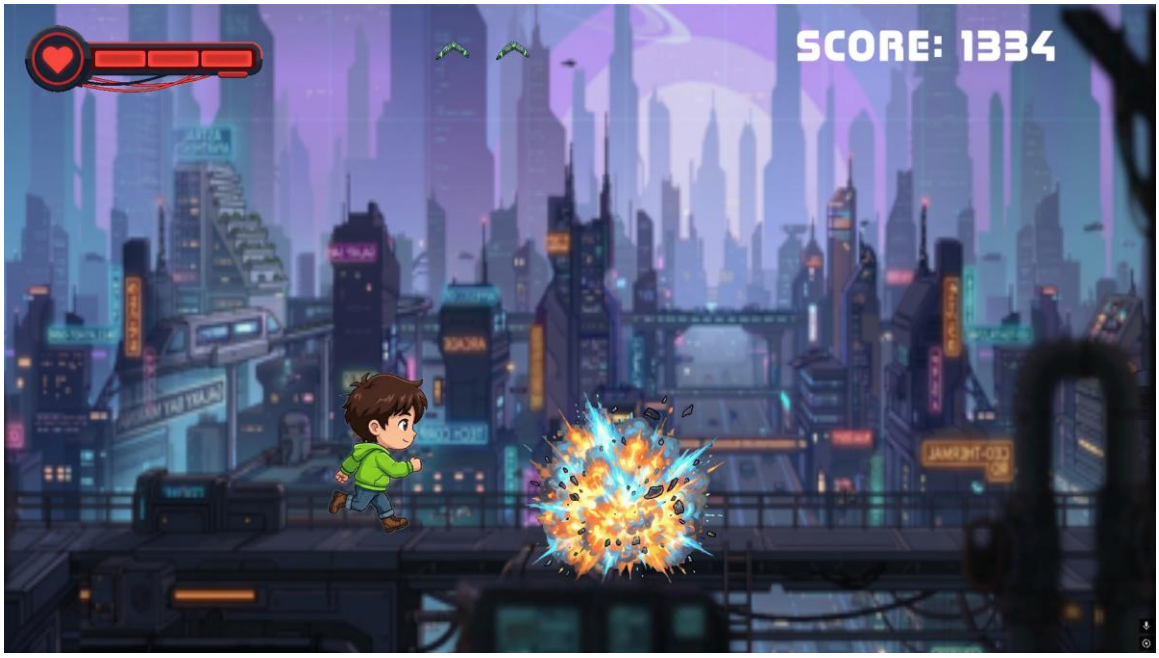
5. Description of the Creative Feature

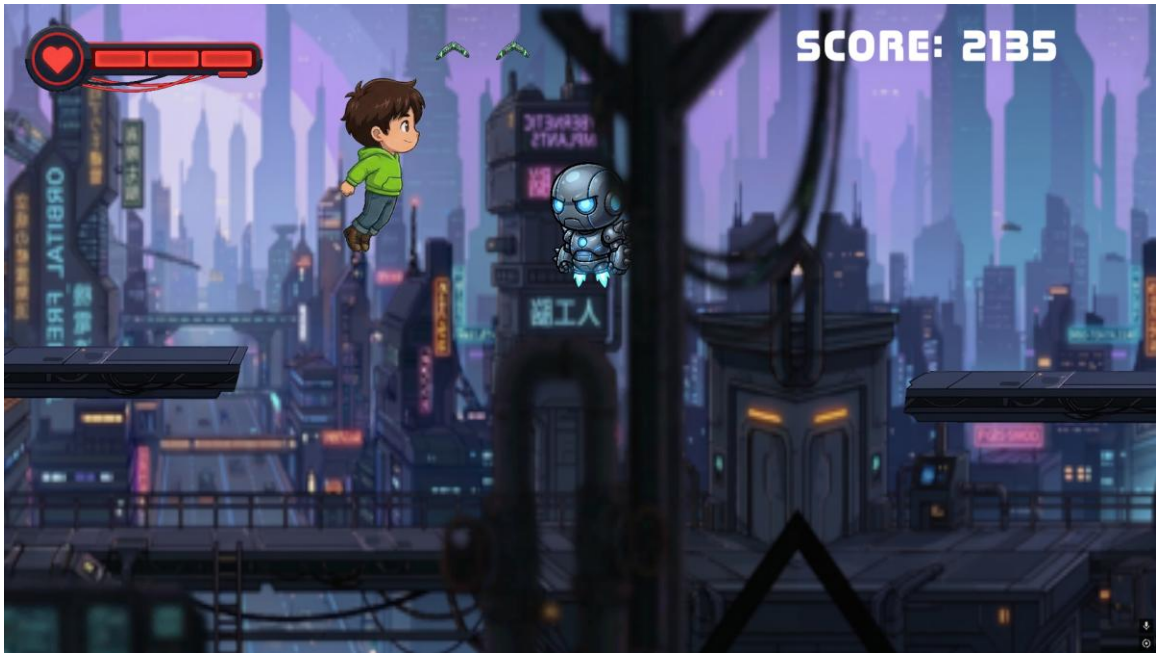
Advanced Boomerang Combat & Parallax Depth:
The primary creative feature is the integration of a physics-based combat system layered over a high-depth visual environment. The player uses a limited boomerang system, tracked via a visual UI array. When thrown, the boomerang utilizes a Rigidbody2D (with 0 gravity scale and continuous collision detection) to fly forward and spin.

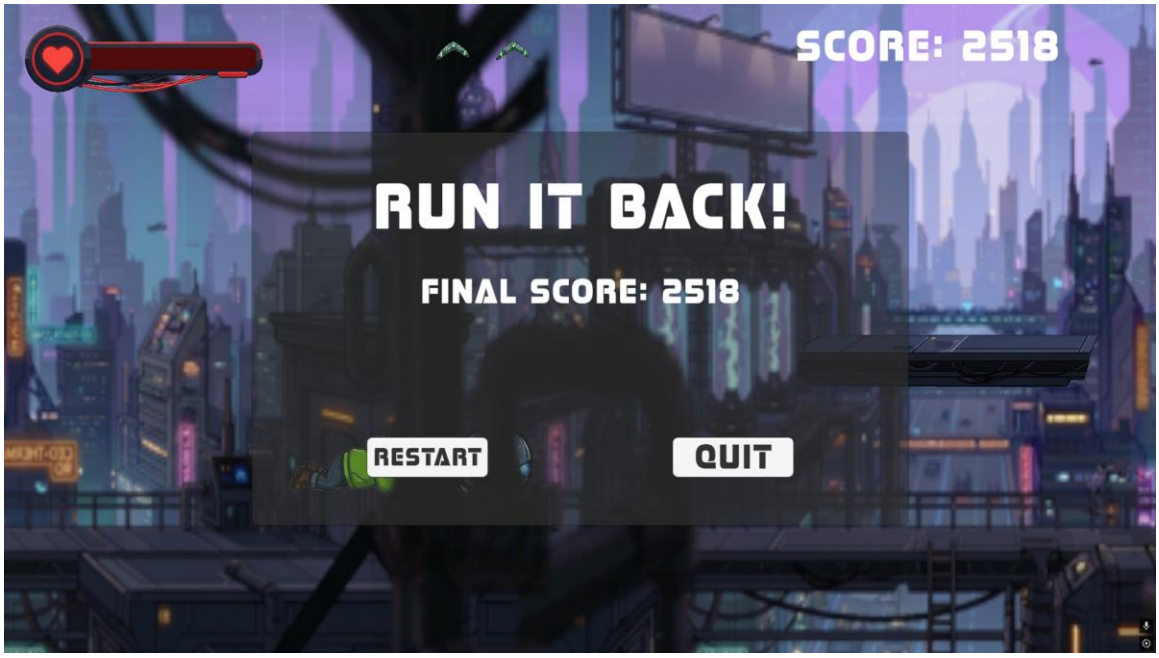
Upon striking a robot enemy, the system instantiates a custom 4-frame explosion animation prefab that inherits the world's movement speed (ObjectMover), ensuring the explosion travels backward with the background rather than freezing statically in the air. Furthermore, a custom transparent Foreground Quad utilizing a Sorting Group forces blurred pipes and wires to render in front of the player and enemies. This foreground moves faster than the distant city background, creating a highly polished, multi-layered 3D parallax depth-of-field effect.

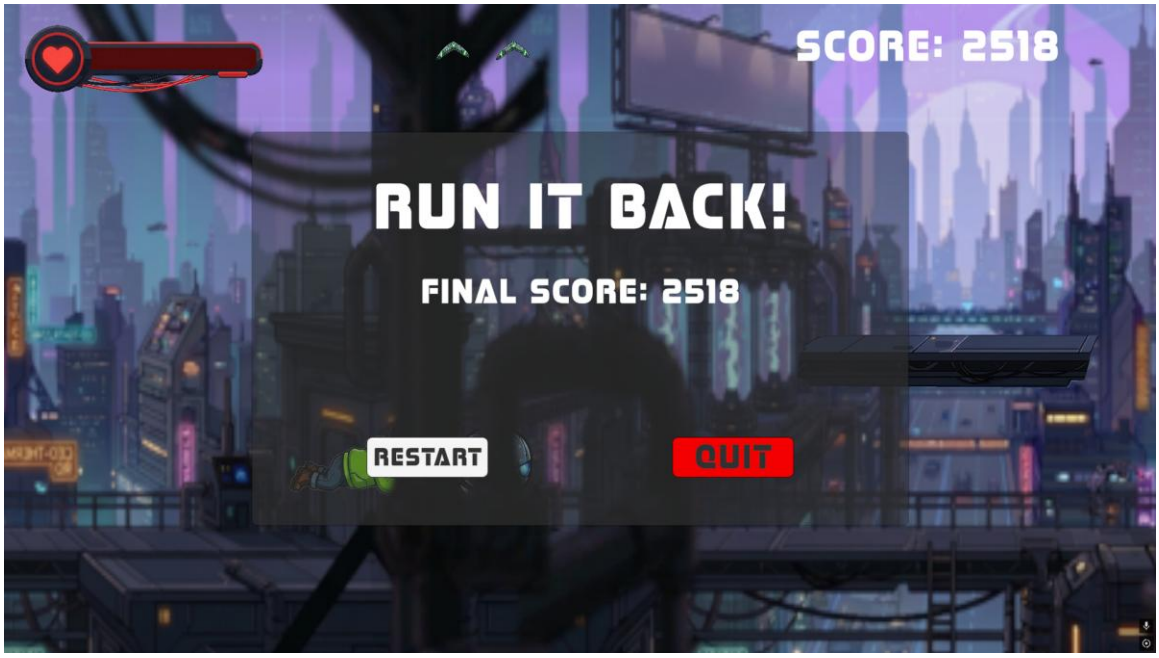
6. Screenshots of Gameplay


SCORE: 0
GO!


SCORE: 1334


SCORE: 2135


SCORE: 2518
RUN IT BACK!
FINAL SCORE: 2518
RESTART
QUIT



7. Credits & External Assets

- Game Engine: Unity 6 (6.3 LTS)
- Visual Assets: Custom Sprites (Player, Robot Enemy, Boomerang, UI Icons)
- Environment Art: Custom Cyberpunk City Background and Blurred Foreground Pipe overlay edited via Photoshop.
- Audio: Custom sound effects for jumping, taking damage, robot destruction, and UI clicks.

8. Video Submission Link

https://drive.google.com/drive/folders/1KAegoypbtSyyaT-6zkNVRcSc7xLDudo3?usp=sharing